

Extreme Temperatures Do Not Deter Modern Electronics

Earlier active or passive air-conditioning and tough verification and qualification processes were needed to maintain electronics used in extreme environments such as mines, spacecrafts, engines and icy mountains. However, advancements in process technology, circuit design and layout techniques have now given rise to electronics that can withstand severe temperatures quite nonchalantly



JANANI GOPALAKRISHNAN VIKRAM

For ages, machines have been put to use in hazardous environments that intimidate humans, and these machines, more often than not, contain electronic components too. In the past, it was a huge challenge for

engineers to make electronics work at temperatures too high or too low (say, less than -55°C or greater than $+125^{\circ}\text{C}$). A lot of extra effort went into active and passive air-conditioning and ventilation systems, to keep the temperature in check, as well as complex verification, characterisation and qualification using lab ovens or freez-

ers, to modify off-the-shelf components and test their behaviour under extreme temperatures. Despite all the efforts, the chances of the system functioning perfectly were still dismal.

"Traditionally, engineers had to rely on active or passive cooling when designing electronics that must function outside normal temperature ranges, but in some applications cooling may not be possible, or it might be more appealing for the electronics to operate hot to improve system reliability and reduce cost. The choice presents challenges that affect many aspects of the electronic system including silicon, packaging, qualification methodology and design techniques," says Somshubhro (Som) Pal Choudhury, managing director, Analog Devices India.

With advances in materials and process technology, circuit design and layout, chip-makers have developed electronic components that can work reliably under extreme-temperature conditions. In fact, players like Analog Devices (ADI) and Texas Instruments (TI) cover the whole signal chain, including sensors, amplifiers and micro-controllers, capable of operating under harsh conditions.

Traditional challenges

Operating electronics at low temperatures is less of a challenge compared to higher ranges. The performance of electronic components is known to be better at lower temperatures. Taking advantage of the same, Kryotech came up with the Cool Athlon range of cen-